Emmanuel Oppong

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Education

Grambling State University Expected May 2028
B.S. in Computer Science *Grambling, LA*
GPA: 3.95/4.0 | **Coursework:** Data Structures & Algorithms, Discrete Structures, Operating Systems, Computer Networks, Programming Language Concepts, Linear Algebra, Probability & Statistics

Experience

ZipMath | **Live demo** Oct 2026 – Present
Founder & Developer *Cross-platform mobile app*

- Founded and built ZipMath, a mobile app that teaches mental-math shortcuts through short lessons and game-style practice, with **26 levels** across Basic, Intermediate, and Advanced tiers.
- Designed the engagement system: **XP, daily streaks, 3 hearts per practice round, and a 1 to 3 star rating** (no mistakes earns 3 stars), to keep learners practicing.
- Built the app in **React Native** and **TypeScript** with a question engine that computes every answer in code, so practice questions are always correct, and animated lesson and quiz flows that save progress on the device.

Open Source Contributor Feb 2026
Apache Kafka, Kubernetes Java Client, PostgreSQL JDBC *GitHub*

- Merged **3 documentation pull requests** into major open source projects, each through the project's review process: Apache Kafka, Kubernetes Java client, and PostgreSQL JDBC.
- Rewrote the **Kafka Streams** guide and the Javadoc of two core classes (3 files, 43 lines) to explain that enabling topology optimizations takes two steps, with a code example and a warning that `build()` without `Properties` silently skips them.
- Wrote 147 lines of release history (v15.0.0 to v25.0.0, including the v20.0.0 breaking changes) for the Kubernetes client, and documented the SCRAM authentication test setup for the PostgreSQL JDBC driver.

Projects

GPU Robot Lab | *Python, Triton, PyTorch, MuJoCo, JAX* | *Live demo*

- Wrote **Triton** matrix-multiply kernels verified against **PyTorch** on an NVIDIA T4. Traced the slow first version to **register spilling** and tuned it **3.4x** faster (1,096 to 3,727 GFLOPS), and found its compiled code uses no tensor cores, the likely cause of the remaining gap.
- Built a **MuJoCo MJX** robot simulation up to **126x** faster than one CPU thread on a GPU (2.75x once collisions are added), with a planner 2 to 26x faster on the GPU than the CPU at the same batch size, and a neural-network controller, trained with a curriculum, that swings up and holds a pendulum.
- Ran controlled experiments: 10 repeats per version (PyTorch itself varied up to 44%), and a one-change-at-a-time test showing the grouped launch order adds 4 to 14%. Backed by **90+ automated tests** in **CI** and an interactive explainer site on GitHub Pages.

InterviewPair | *Java, Spring Boot, PostgreSQL, React, TypeScript, Docker*

- Built the core of an app that pairs students for mock technical interviews: a **Spring Boot** API with **JWT** login, hashed passwords and a tested matching rule, and a **React** frontend, in one **Docker** image.
- Wrote 30+ automated tests, including expired, tampered and wrongly signed login tokens, and **CI** on GitHub Actions that builds the Docker image on every push.

Stock Price Prediction (AI4ALL Ignite, team project) | *Python, FinBERT, Scikit-learn*

- Contributed to a 3-person team project that predicts next-day stock decreases from financial news headlines, using **FinBERT** sentiment features and 20M+ stock price records. The **Random Forest** reached 54.4% accuracy against 50.8% for the **Logistic Regression** baseline on 64,599 held-out headlines.
- Wrote the evaluation script: **5-fold stratified cross-validation** on a 50,000-article sample, with classification reports and confusion matrices for both models.

Technical Skills

Languages: Python, TypeScript, JavaScript
Frameworks & Libraries: React, Next.js, Tailwind CSS, Scikit-learn
Tools: Git, GitHub, Supabase

Leadership & Awards

ColorStack Grambling, Academic Chair Aug 2026 – Present
Lead weekly data structures and algorithms workshops for **90+ members**.
AWS Student Builder Group Leader, Grambling State University Starts Nov 2026
Awards: 2x Dean's List; 4x Academic Excellence Scholarship